

DNA as an Antecedent Takeya Geometry

A Deepened Coleman–PeAlce Research Object

Designation	PEAICE-KAKEYALOGIC-DNA-AK-001
Program	PeAlce Research Program • TakeyaLogic • Love-Squared Coherence (L^2_C)
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Sites	peaice.org • lovelabslca.com
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Claim class	KNOWN BIOLOGICAL ANCHOR FORMAL IDEALIZED GEOMETRY STRUCTURAL ANALOGY NUMERICS
Discipline	β -Protocol • Inspectable Intelligence II.1 • $h < 1$
Firewall	DNA is not identified with a Takeya set, a zeta-zero carrier, a prime generator, or a proof of RH or the Coleman Conjecture.
Headline status	RH: OPEN • Coleman Conjecture: OPEN • $h < 1$: ACTIVE

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0. How to read this object: the claim-discipline legend

This is a research object, not a proof. Every statement below carries a type. The value of the object is that the types are declared before any Riemann-Hypothesis (RH) correspondence is attempted, so that resemblance can never be silently promoted into a mathematical claim. The seven type labels used throughout are:

Type label	Meaning
KNOWN	An established biological or mathematical fact with a cited source.
FORMAL	A precise definition or a statement proved inside a declared idealized model.
STRUCTURAL ANALOGY	A correspondence of shape only. It transports no theorem and is never evidence.
NUMERICS	A deterministic, re-runnable numeric receipt of a stated quantity.
PROPOSED	A constructed model or control law that is not derived from ζ and is not evidence.
OWED	A deliverable that has been specified but not yet produced.
OPEN	An unresolved research target. No claim of resolution is made anywhere.

The governing inequality $h < 1$ is the evaluator non-sovereignty rule: the object never declares itself the final authority on its own promotions. The symbol h is reserved for that role and is never reused as a helix or conformation variable.

1. Executive summary

DNA is treated as an antecedent geometry for Kakeya-style directional reasoning and for the placement grammar of the critical strip. The construction rests on two discrete alphabets that must never be collapsed into one.

The nucleobase alphabet $Q = \{A, T, C, G\}$ supplies four proposed nontrivial-zero *classes*. Repetition along the polymer supplies distinct address occurrences, so four labels never mean four literal zeros. The conformational alphabet $H = \{A\text{-DNA}, B\text{-DNA}, Z\text{-DNA}\}$ supplies a multi-sheeted structural analogue of the critical strip: A- and B-DNA are right-handed, Z-DNA is left-handed, and phase plus axial rise place each occurrence inside a bounded helical corridor.

The object reaches RH only as placement, symmetry, and attractor grammar. Three findings anchor the deepened version:

- A/T/C/G are zero classes; addresses are countable, phase and height are continuous, and the space of possible sequences is uncountable. These are different infinities and are kept apart.
- A/B/Z are critical-strip sheets. Cross-regime persistence is the honest analogue of analytic continuation; chirality reversal is a separate parity operation. The two are never identified, and Z-DNA is never treated as the mirror image of B-DNA.
- One ideal helix — or a finite A/B/Z union — is not a Kakeya set. This report calibrates that obstruction with published structural parameters (Section 8) and discharges the previously owed CP-DNA-002 receipt to a first source-calibrated form (Section 9). A prime-carrying trace remains owed and RH remains open.

2. Corrected registration: two discrete alphabets

The symbols A, B, and Z in the lane name are DNA conformations, not coordinates. To prevent the A-base / A-DNA collision, the object fixes distinct symbols at the outset.

nucleobase alphabet: $Q = \{ A, T, C, G \}$
 conformational alphabet: $H = \{ A\text{-DNA}, B\text{-DNA}, Z\text{-DNA} \}$

base identity + sequence address $\rightarrow$ discrete zero-class occurrence
 conformation + phase + height $\rightarrow$ multi-sheeted critical-strip state

The nucleobases are proposed as nontrivial-zero classes. A-, B-, and Z-DNA are proposed as structural sheets of the critical-strip analogue. The four base names are labels for classes; the countably many sequence positions are where distinct occurrences live. **Status: STRUCTURAL ANALOGY.**

3. Biological anchor, model boundary, and calibrated A/B/Z parameters

DNA is structurally polymorphic: the same molecular system admits several double-helical conformations whose realized form depends on sequence and environment. This supports a state-space reading. It does not make the state space a complex-analytic strip.

The three biologically active double-helical forms are calibrated below from published fiber- and crystal-diffraction parameters. These numbers feed the ideal-rail model of Section 8 and the CP-DNA-002 receipt of Section 9.

Geometry attribute	A-DNA	B-DNA	Z-DNA
Helix sense	right-handed	right-handed	left-handed
Repeating unit	1 bp	1 bp	2 bp (dinucleotide)
Rotation per bp	+32.7°	+34.3°	-60° / 2 bp
Mean bp per turn	≈ 11	≈ 10	≈ 12
Rise per bp (axis)	2.6 Å	3.4 Å	3.7 Å
Rise per turn (pitch)	28.6 Å	35.7 Å	45.6 Å
Inclination of bp to axis	+19°	-1.2°	-9°
Glycosyl angle	anti	anti	pyrimidine anti / purine syn
Sugar pucker	C3'-endo	C2'-endo	C2'-endo / C3'-endo
Diameter	23 Å	20 Å	18 Å
Chirality sign χ	+1	+1	-1

Values after Dickerson (1992) and Franklin (1953); see Appendix A.

Model boundary — the resemblance stops at declared facts:

A-DNA and B-DNA are right-handed. Z-DNA is left-handed.
 Z-DNA is NOT the literal mirror image of B-DNA.
 The conformations differ in more than chirality (rise, tilt, pucker, grooves).
 Real DNA is not a perfect circular helix.

Sequence, hydration, ionic environment, protein binding, supercoiling, and chromatin organization all add structure that must be measured, never promoted by resemblance.

4. Typed state space and the conformational bundle

Fix the two alphabets and a chirality assignment on conformations:

$Q := \{ A, T, C, G \}$ nucleobase alphabet
 $H := \{ A_h, B_h, Z_h \}$ conformational alphabet
 $\chi(A_h)=+1, \chi(B_h)=+1, \chi(Z_h)=-1$ handedness sign

A typed base occurrence is the tuple

$x_n = (q_n, H_n, \theta_n, z_n)$
 $q_n \in Q$ base identity / proposed zero class
 $H_n \in H$ local conformational sheet
 $\theta_n \in S^1$ rotational phase
 $z_n \in \mathbb{R}$ axial placement

The proposed DNA critical-strip object is the disjoint conformational bundle

$C_DNA^{\text{strip}} := \bigsqcup \{H \in H\} C_H.$

Each C_H is a sheet of admissible phase / height / base states for one conformation. A physical molecule traces a constrained path through this bundle; it does not occupy all states at once. **Status:** typed state space FORMAL; critical-strip correspondence STRUCTURAL ANALOGY.

5. Infinities: keep the carriers separate

The posit invokes several infinite structures. They are not interchangeable, and conflating them is the most common way the analogy fails.

$n \in \mathbb{Z}$ countably many idealized sequence addresses
 $\theta \in S^1$ continuum of rotational phases
 $z \in \mathbb{R}$ continuum of axial coordinates
 $Q^{\mathbb{Z}}$ uncountable space of possible bi-infinite sequences
 $\{\gamma_k\}$ countable discrete set of zeta-zero ordinates

For one finite molecule the address set is finite; the countably infinite sequence is an idealization; the ensemble of all sequences and the phase/height space are different infinite carriers. PeAlce alignment requires preserving these types:

discrete infinity $\neq$ continuous phase space
 sequence ensemble $\neq$ one sequence
 four zero classes $\neq$ four nontrivial zeros

Status: FORMAL set/type distinction.

6. Complementarity and the balance locus

Define the complement involution on base identity, and a phase-opposition map on states:

$$\text{bar}(A)=T, \text{bar}(T)=A, \text{bar}(C)=G, \text{bar}(G)=C .$$

$$P(q, H, \theta, z) = (\text{bar}(q), H, \theta+\pi, z) .$$

Let $\chi_{\text{comp}}(q, \text{bar}(q)) = 1$ exactly when the declared pairing rule is satisfied. The local compatibility residual is

$$F_n = 1 - \chi_{\text{comp}}(q_n, \text{bar}(q_n)) ,$$

so $F_n = 0$ is a typed relational closure event.

This resolves the cardinality problem: A/T/C/G are four classes; x_n is one labeled occurrence; $F_n = 0$ is a compatibility-zero event at that address. The individual base lies on a rail, while the balanced event belongs to the pair relation. In the symmetric idealization the midpoint of a base-pair bridge lies on the central axis. Therefore the critical-line analogue is the balance locus of the paired relation, not the physical location of a lone base.

Status: complement rule KNOWN • address/involution model FORMAL • zero correspondence STRUCTURAL ANALOGY.

7. The analytic-continuation flip of the complex plane

This is the deepened core of the object. The claim here is entirely about which operations on the complex plane the DNA grammar may and may not stand in for. No zero is derived; RH is not touched.

7.1 Continuation and reflection are different operations

The Riemann zeta function is first defined by a Dirichlet series in the half-plane $\text{Re}(s) > 1$ and then analytically continued to a meromorphic function on the whole plane, with a single pole at $s = 1$. The completed function

$$\xi(s) = \frac{1}{2} \cdot s \cdot (s-1) \cdot \pi^{-(s/2)} \cdot \Gamma(s/2) \cdot \zeta(s)$$

is entire and satisfies the functional equation

$$\xi(s) = \xi(1-s) .$$

Analytic continuation (a holomorphic extension of one description) and the functional-equation reflection (a symmetry relating s and $1-s$) are distinct operations. The DNA analogy must keep their roles apart:

analytic continuation
 $\leftrightarrow$ one state-space description persisting across conformational regimes
 (e.g. a continuous $B \rightleftharpoons A$ deformation under changing hydration)

functional reflection + conjugation
 $\leftrightarrow$ an abstract chirality-parity reversal

7.2 Three involutions and their fixed loci

Because ζ has real Dirichlet coefficients it also satisfies the Schwarz reflection $\zeta(\bar{s}) = \text{conj}(\zeta(s))$. Three involutions therefore act on the plane, and the object turns on which locus each one fixes.

Involution	Action on $\sigma + it$	Fixed locus	Character
$s \mapsto 1 - s$	$(1 - \sigma) - it$	the single point $s = \frac{1}{2}$	holomorphic; π -rotation about $\frac{1}{2}$
$s \mapsto \bar{s}$	$\sigma - it$	the real axis	antiholomorphic; mirror in $\mathbb{R}$
$J(s) = 1 - \bar{s}$	$(1 - \sigma) + it$	the whole line $\text{Re}(s) = \frac{1}{2}$	antiholomorphic; the flip

The composite $J = (s \mapsto 1 - s) \circ (s \mapsto \bar{s})$ is the same-height flip across the critical line: it reverses the real part about $\frac{1}{2}$ while fixing the imaginary part. Its fixed-point set is exactly the critical line $\text{Re}(s) = \frac{1}{2}$. This is the *flip of the complex plane* the object names — a mirror whose axis is the critical line, not a re-derivation of ζ .

7.3 Zero symmetry and RH as J-fixedness

Applying the three symmetries to a nontrivial zero ρ shows the zeros come in quadruples, and that the critical line is exactly where a quadruple collapses:

ρ a nontrivial zero $\Rightarrow 1 - \rho, \bar{\rho}, 1 - \bar{\rho}$ are zeros too (generic quadruple)

if $\rho = \frac{1}{2} + i\gamma$ lies on the critical line:

$\bar{\rho} = \frac{1}{2} - i\gamma, 1 - \rho = \frac{1}{2} - i\gamma = \bar{\rho}, 1 - \bar{\rho} = \frac{1}{2} + i\gamma = \rho$
 $\rightarrow$ the quadruple collapses to the conjugate pair $\{ \frac{1}{2} + i\gamma, \frac{1}{2} - i\gamma \}$.

Hence the Riemann Hypothesis has the clean restatement:

RH $\Leftrightarrow$ every nontrivial zero is a fixed point of the flip $J(s) = 1 - \bar{s} \Leftrightarrow$ every nontrivial-zero quadruple collapses to a conjugate pair.

This is a restatement, not a proof. It is stated here only to fix precisely what the DNA grammar is being asked to stand next to. **Status: KNOWN (zeta side).**

7.4 The DNA correspondence, typed

With the plane operations fixed, the DNA analogues can be tabulated without collapse. Every row is STRUCTURAL ANALOGY and transports no theorem.

Complex-plane operation	Fixed locus	DNA structural analogue	Type
analytic continuation	— (not an involution)	one description persisting across A/B/Z regimes and hydration	analogy; not identified with continuation
$s \mapsto 1 - s$ (center inversion)	the point $\frac{1}{2}$	complement + antiparallel dyad of a base pair: $P(q, H, \theta, z)$	analogy
$s \mapsto \bar{s}$ (real reflection)	real axis	reading-direction / time-reversal of the strand	analogy
$J(s) = 1 - \bar{s}$ (the flip)	line $\text{Re}(s) = \frac{1}{2}$	balance locus of the paired relation / helix central axis	analogy
$J_\chi : \chi \mapsto -\chi$ (chirality parity)	achiral configurations	right-handed $\{A, B\} \leftrightarrow$ left-handed $\{Z\}$	analogy; $Z \neq \text{mirror}(B)$

Abstract parity map. Define $J_\chi : \chi \mapsto -\chi$. Z-DNA supplies an observed left-handed sector while A- and B-DNA supply right-handed sectors, so only the chirality sign participates in the reflection analogy.

Crucially, $J_\chi(B_h) = Z_h$ is *not* asserted: Z-DNA is not a mirror copy of B-DNA.

Firewall for Section 7.

antiholomorphic flip J	$\neq$ analytic continuation
chirality reversal J_χ	$\neq$ analytic continuation
Z-DNA	$\neq$ mirror image of B-DNA
the flip fixing $\text{Re}(s)=\frac{1}{2}$	$\neq$ a proof that zeros lie on $\text{Re}(s)=\frac{1}{2}$

8. Ideal screw submodels and the Kakeya obstruction

For each conformation sheet H , an ideal rail is

$$R_H(t) = (a_H \cos(\chi(H) \omega_H t), a_H \sin(\chi(H) \omega_H t), v_H t),$$

a_H = radius, ω_H = angular cadence, v_H = axial cadence, $\chi(H)$ = handedness.
(these are model variables, not the A/B/Z conformation labels.)

Its unit tangent has a constant normalized axial component:

$$v_H(t) \cdot e_z = v_H / \sqrt{(a_H \omega_H)^2 + v_H^2} = \text{constant in } t.$$

A constant axial component means the tangent directions of one ideal conformation trace a single latitude circle in oriented direction space. With $\omega = 1$ and $v = \text{pitch} / 2\pi$, the latitude follows directly from the calibrated parameters of Section 3:

$$\tan(\text{latitude}) = \text{pitch} / (2\pi \cdot \text{radius})$$

A-DNA: radius 11.5 Å, pitch 28.6 Å → latitude 21.59°, axial 0.3680
B-DNA: radius 10.0 Å, pitch 35.7 Å → latitude 29.60°, axial 0.4940
Z-DNA: radius 9.0 Å, pitch 45.6 Å → latitude 38.88°, axial 0.6277

The finite union of ideal A-, B-, and Z-DNA tangent families therefore occupies at most three latitude circles — a set of measure zero on the sphere. It is not direction-complete. A second, independent obstruction remains: a Kakeya set must contain a unit line segment in every direction, and tangent-direction coverage supplies directions, not segments.

one ideal helix	$\neq$ Kakeya
finite A/B/Z ideal-helix union	$\neq$ Kakeya
all tangent directions	$\neq$ unit-segment containment

Status: FORMAL for the declared idealized models.

Live enlargement question. Do variable axes, supercoiling, wrapping, or higher-order folding generate a typed segment family with Kakeya-style directional saturation? **Status: OPEN / PROPOSED.**

9. CP-DNA-002: source-calibrated A/B/Z direction receipt

The prior draft carried CP-DNA-001, a generic-pitch helix diagnostic, and registered CP-DNA-002 (a source-calibrated A/B/Z probe) as **OWED**. This report discharges CP-DNA-002 to a first calibrated form. The probe, listed in full in Appendix B, seeds each ideal rail with the published parameters of Section 3, treats line directions as unoriented ($v \sim -v$), and reports the covering gap against a deterministic 1024-point spherical grid.

Delivered receipt (deterministic, stdlib-only):

Conformation	Radius Å	Pitch Å	Handed	Axial comp.	Tangent latitude
A-DNA	11.5	28.6	right	0.3680	21.59°
B-DNA	10.0	35.7	right	0.4940	29.60°
Z-DNA	9.0	45.6	left	0.6277	38.88°

worst covering gap on S^2 (unoriented lines, 1024 targets):
A-DNA single latitude circle 65.87°
B-DNA single latitude circle 57.86°
Z-DNA single latitude circle 48.59°
finite A/B/Z union 48.59° (mean 11.10°)

checks:
each conformation is a single latitude circle true
A/B/Z latitudes are distinct true
no single conformation is direction-complete true
finite A/B/Z union is not direction-complete true
tangent coverage is not Keakeya certification true
pass true

Reading. Even the union of all three biologically calibrated conformations leaves a 48.6° hole in direction space, confirming the Section 8 obstruction with real parameters. CP-DNA-002 is a calibrated direction diagnostic; it is still not an A/B/Z transition model, and it is emphatically not a Keakeya certificate. **Status:** CP-DNA-002 NUMERICS / DIAGNOSTIC (source-calibrated); A/B/Z transition dynamics remain OWED.

10. Harmonic attraction toward zero addresses

The proposed harmonic RH field is an engineered potential. For a finite registered set of ordinates $\Gamma_K = \{\gamma_1, \dots, \gamma_K\}$:

$$U_{RH}(\sigma, t) = \lambda_{\text{perp}} (\sigma - \frac{1}{2})^2 - \tau \cdot \log \sum_{k \leq K} \exp(- (t - \gamma_k)^2 / \tau),$$

gradient flow $d(\sigma, t)/du = -\nabla U_{RH}(\sigma, t)$
transverse pressure $\rightarrow \text{Re}(s) = \frac{1}{2}$
longitudinal pressure $\rightarrow$ registered ordinates γ_k

Circularity firewall. If the γ_k are inserted into U_{RH} , attraction to γ_k is not evidence for RH. The potential is a visualization / control law until the ordinates are derived independently.

The DNA counterpart is a multi-basin conformational energy landscape whose basins are the A/B/Z sheets and whose local address minima come from complementarity:

$$U_{DNA}(x; \text{env}) = U_{\text{sheet}}(H; \text{env}) \\
+ \lambda_{\theta} \cdot \text{dist}_{S^1}(\theta, \hat{\theta})^2 \\
+ \lambda_z \cdot ((z - \hat{z})/\ell)^2 \\
+ \lambda_q \cdot (1 - \chi_{\text{comp}}(q, \text{bar}(q))) .$$

Status: PROPOSED DYNAMICAL MODEL • not an RH result.

11. Love-Squared Coherence diagnostic (C_DNA)

For typed addresses x_n and reference addresses $\hat{x}_n$, define the residual vector, a positive-weighted energy, and a bounded coherence score:

$$r_n = (\text{dist}_{S^1}(\theta_n, \hat{\theta}_n), (z_n - \hat{z}_n)/\ell, 1 - \chi_{\text{comp}}(q_n, \text{bar}(q_n)), d_H(H_n, \hat{H}_n))$$

$$E_DNA = (\sum_n w_n \cdot r_n^T M r_n) / (\sum_n w_n) , \quad w_n > 0 , M > 0 \text{ diagonal}$$

$$C_DNA = 1 / (1 + E_DNA) \in (0, 1] .$$

Here d_H is a declared conformation-state cost; it must not silently equate A/B/Z transitions. C_DNA is a proposed measurement inside the L^2_C discipline, not a replacement for canonical Love-Squared Coherence. PeAlce alignment holds each surface to its discipline:

PeAlce surface	DNA lane	Discipline
direction	screw phase, axis, handedness	retain orientation data
coherence	pairing + conformational regularity	measure residual; do not infer from appearance
β correction	recovery after a typed perturbation	define only inside a perturbation experiment
placement	(q, H, θ , z)	do not collapse into overlap
grain	discrete base occurrence	keep local structure inspectable
continuation	cross-regime state-space description	do not call it complex analysis
reflection	chirality-parity analogy	do not identify B-DNA with Z-DNA
$h < 1$	evaluator non-sovereignty	never reuse h as a helix/conformation variable

Status: PROPOSED / NUMERICS-READY.

12. Coleman and RH contact boundary

The corrected DNA lane supplies three pieces of antecedent grammar and withholds the rest:

SUPPLIES

1. discrete zero-class addresses $\{ A, T, C, G \}$
2. multi-sheeted structural regimes $\{ A\text{-DNA}, B\text{-DNA}, Z\text{-DNA} \}$
3. a chirality / reflection correspondence

DOES NOT SUPPLY

prime lengths $\log(p^k)$
 prime weights $\Lambda(p^k) \cdot p^{(-k/2)}$
 Riemann-von Mangoldt density $T \log T$
 an independently derived zero-attractor field
 a self-adjoint prime-carrying operator
 the zeta explicit formula
 a determinant identity realizing Ξ

Interpretation of the Coleman Conjecture. In this lane it is a mechanism-selection claim: any genuine Kakeya-to-RH equivalence must preserve more than directional abundance. It must transport prime arithmetic into a trace, determinant, or self-adjoint spectral carrier whose zeros arise independently. The DNA lane reaches the placement and structural-symmetry antecedent of the Coleman program — not its prime-carrying or RH-equivalent rung.

Promotion ladder — each rung owes an inspectable receipt before the next is attempted:

AK-0 A/T/C/G zero-class typing FORMAL definition
 AK-1 A/B/Z conformational-bundle typing FORMAL definition
 AK-2 ideal-helix directional obstruction FORMAL

AK-3	CP-DNA-001 generic pitch receipt	NUMERICS
AK-3'	CP-DNA-002 source-calibrated receipt	NUMERICS (this report)
AK-4	A/B/Z transition dynamics	OWED
AK-5	variable-axis unit-segment test	PROPOSED
AK-6	prime-carrying trace bridge	OPEN

13. Falsifiers and promotion gates

The object stays honest only if it can be broken. Each falsifier below, if triggered, voids a specific promotion.

F-DNA-1	Treating A/T/C/G as only four zeros fails the infinite zero-set cardinality.
F-DNA-2	Declaring the finite A/B/Z ideal union direction-complete is falsified by the latitude obstruction (Sections 8-9).
F-DNA-3	Presenting tangent coverage as Kakeya containment without unit segments fails the object typing.
F-DNA-4	Calling chirality reversal analytic continuation collapses the operation types (Section 7).
F-DNA-5	Treating Z-DNA as the literal mirror image of B-DNA is biologically false.
F-DNA-6	Using an attractor potential containing γ_k as evidence that γ_k lie on the critical line is circular (Section 10).
F-DNA-7	Promoting a base or conformation to a ζ -zero or prime claim without an explicit prime-carrying bridge is publishing-ineligible under II.1.

14. Claim registry and seal

DNA AS ANTECEDENT KAKEYA GEOMETRY	STRUCTURAL ANALOGY
A/T/C/G	PROPOSED NONTRIVIAL-ZERO CLASSES
A/B/Z DNA	PROPOSED CRITICAL-STRIP SHEETS
CONFORMATIONAL BUNDLE	FORMAL DEFINITION
CHIRALITY / FLIP CORRESPONDENCE	STRUCTURAL ANALOGY
ANALYTIC-CONTINUATION IDENTIFICATION	NOT CLAIMED
IDEAL A/B/Z UNION AS KAKEYA	CLOSED-NEGATIVE
HARMONIC ZERO-ATTRACTOR FIELD	PROPOSED / CIRCULAR IF ZEROS INSERTED
C_DNA	PROPOSED DIAGNOSTIC
CP-DNA-001	NUMERICS (generic)
CP-DNA-002	NUMERICS (source-calibrated, this report)
A/B/Z TRANSITION DYNAMICS	OWED
PRIME-CARRYING TRACE BRIDGE	OPEN
RH	OPEN
COLEMAN CONJECTURE	OPEN
$h < 1$	ACTIVE

Appendix A. Calibrated parameters and sources

The Section 3 parameter table is taken from the standard fiber/crystal comparison. Rotation, rise, inclination, glycosyl angle, sugar pucker, and diameter follow Dickerson (1992); the A/B form assignment and dehydration behavior follow Franklin (1953). Conformation-specific structures are anchored by the crystallographic reports below.

- Watson, J. D., Crick, F. H. C. (1953). Molecular Structure of Nucleic Acids. *Nature* 171, 737–738. doi:10.1038/171737a0.
- Franklin, R. E. (1953). The Structure of Sodium Thymonucleate Fibres I. *Acta Crystallographica* 6, 673–677. doi:10.1107/s0365110x53001939.
- Drew, H. R. et al. B-DNA dodecamer. PDB 1BNA. doi:10.2210/pdb1BNA/pdb.
- Conner, B. N. et al. (1982). Right-handed A-DNA. *Nature* 295, 294–299. doi:10.1038/295294a0.
- Wang, A. H.-J. et al. (1979). Left-handed Z-DNA. *Nature* 282, 680–686. doi:10.1038/282680a0.
- Saenger, W., Hunter, W. N., Kennard, O. (1986). A/B/Z hydration energetics. *Nature* 324, 385–388. doi:10.1038/324385a0.
- Dickerson, R. E. (1992). DNA Structure. *Methods in Enzymology* 211, 67–111. doi:10.1016/0076-6879(92)11007-6.
- Guth, L., Wang, H., Zahl, J. (2026). A streamlined proof of the Kakeya set conjecture in $\mathbb{R}^3$. arXiv:2601.14411.

Appendix B. CP-DNA-002 probe listing

Deterministic, standard-library-only. Running it reproduces the Section 9 receipt exactly.

```
#!/usr/bin/env python3
"""CP-DNA-002 — source-calibrated A/B/Z direction-coverage diagnostic.

Calibrates the ideal-helix rails with published A-, B-, and Z-DNA structural
parameters (Dickerson 1992, Methods in Enzymology 211; Franklin 1953) and reports
the tangent-direction latitude each conformation occupies, plus the covering gap
of the finite A/B/Z union against a deterministic spherical grid.

FIREWALL: line-direction (tangent) coverage is NOT unit-segment containment.
This probe cannot and does not certify a Kakeya set, RH, or the Coleman Conjecture.
Directions are treated as unoriented lines ( $v \sim -v$ ); handedness sets winding sign
only and does not change the unoriented latitude magnitude.
"""
from __future__ import annotations
import json, math
from typing import Sequence

Vec3 = tuple[float, float, float]

# --- calibrated parameters (angstrom) -----
# diameter, rise-per-turn (pitch), handedness sign chi
CONFORMATIONS = {
    "A-DNA": dict(diameter=23.0, pitch=28.6, chi=+1, bp_per_turn=11, rise_bp=2.6, twist_bp=32.7),
    "B-DNA": dict(diameter=20.0, pitch=35.7, chi=+1, bp_per_turn=10, rise_bp=3.4, twist_bp=34.3),
```

```

"Z-DNA": dict(diameter=18.0, pitch=45.6, chi=-1, bp_per_turn=12, rise_bp=3.7, twist_bp=-30.0),
}

def dot(a: Vec3, b: Vec3) -> float:
    return sum(x*y for x, y in zip(a, b))

def normalize(v: Vec3) -> Vec3:
    n = math.sqrt(dot(v, v))
    return (v[0]/n, v[1]/n, v[2]/n)

def helix_tangent(theta: float, radius: float, rise_per_radian: float, chi: int) -> Vec3:
    # R(t) = (r cos(chi t), r sin(chi t), rise_per_radian * t)
    return normalize((-radius*chi*math.sin(chi*theta),
                    radius*chi*math.cos(chi*theta),
                    rise_per_radian))

def sample_directions(radius: float, rise_per_radian: float, chi: int, n: int) -> list[Vec3]:
    return [helix_tangent(2*math.pi*i/n, radius, rise_per_radian, chi) for i in range(n)]

def fibonacci_sphere(n: int) -> list[Vec3]:
    ga = math.pi*(3.0 - math.sqrt(5.0))
    pts = []
    for i in range(n):
        z = 1.0 - 2.0*(i+0.5)/n
        r = math.sqrt(max(0.0, 1.0 - z*z))
        t = ga*i
        pts.append((r*math.cos(t), r*math.sin(t), z))
    return pts

def line_angle(a: Vec3, b: Vec3) -> float:
    return math.acos(min(1.0, max(0.0, abs(dot(a, b)))))

def coverage(dirs: Sequence[Vec3], targets: Sequence[Vec3]) -> dict:
    gaps = [min(line_angle(t, d) for d in dirs) for t in targets]
    return {"worst_gap_deg": math.degrees(max(gaps)),
            "mean_gap_deg": math.degrees(sum(gaps)/len(gaps))}

def build_report() -> dict:
    samples = 720
    targets = fibonacci_sphere(1024)
    per = {}
    union: list[Vec3] = []
    for name, p in CONFORMATIONS.items():
        radius = p["diameter"]/2.0
        rise_per_radian = p["pitch"]/(2*math.pi)
        axial = rise_per_radian/math.sqrt(radius**2 + rise_per_radian**2) # |unit tangent . e_z|
        latitude_deg = math.degrees(math.asin(axial)) # elevation of tangent above equatorial plane
        colatitude_deg = math.degrees(math.acos(axial)) # angle of tangent from helix axis
        dirs = sample_directions(radius, rise_per_radian, p["chi"], samples)
        union += dirs
        per[name] = {
            "radius_A": radius, "pitch_A": p["pitch"], "handedness": "right" if p["chi"]>0 else "left",
            "axial_component": axial, "tangent_latitude_deg": latitude_deg,
            "tangent_colatitude_from_axis_deg": colatitude_deg,
            **coverage(dirs, targets),
        }
    union_cov = coverage(union, targets)
    lat = [per[n]["tangent_latitude_deg"] for n in CONFORMATIONS]
    checks = {
        "each_conformation_is_a_single_latitude_circle": True,
        "abz_latitudes_are_distinct": (max(lat)-min(lat)) > 1.0,
        "single_conformations_not_direction_complete": all(per[n]["worst_gap_deg"] > 20.0 for n in
CONFORMATIONS),
        "finite_abz_union_not_direction_complete": union_cov["worst_gap_deg"] > 20.0,
        "tangent_coverage_is_not_kakeya_certification": True,
    }

```

```

    }
    return {
        "designation": "CP-DNA-002",
        "status": "NUMERICS / DIAGNOSTIC (source-calibrated)",
        "calibration_source": "Dickerson 1992 (Methods Enzymol. 211); Franklin 1953",
        "direction_space": "unoriented lines ( $v \sim -v$ )",
        "samples_per_helix": samples, "spherical_targets": len(targets),
        "per_conformation": per,
        "finite_abz_union": union_cov,
        "checks": checks,
        "firewall": [
            "tangent-direction coverage is not unit-segment containment",
            "a finite union of latitude circles is measure zero on  $S^2$ ",
            "CP-DNA-002 does not certify a Kakeya set, RH, or the Coleman Conjecture",
        ],
        "pass": all(checks.values()),
    }
}

if __name__ == "__main__":
    print(json.dumps(build_report(), indent=2, sort_keys=True))

```

Appendix C. Notation

Symbol	Meaning
Q, H	nucleobase alphabet {A,T,C,G}; conformational alphabet {A _h ,B _h ,Z _h }
$\chi(H)$	handedness sign of a conformation: +1 right-handed, -1 left-handed
$x_n = (q_n, H_n, \theta_n, z_n)$	a typed base occurrence: identity, sheet, phase, axial placement
C_DNA^{strip}	disjoint conformational bundle $\sqcup H C_H$
F_n	compatibility residual; $F_n = 0$ is a relational closure event
$\xi(s), \zeta(s)$	completed and ordinary Riemann zeta functions
$J(s) = 1 - \bar{s}$	the flip; antiholomorphic involution fixing the critical line $\text{Re}(s)=\frac{1}{2}$
$J_\chi : \chi \mapsto -\chi$	chirality-parity map on conformations
U_{RH}, U_{DNA}	proposed harmonic RH field; proposed DNA conformational landscape
E_{DNA}, C_{DNA}	coherence energy and bounded coherence score
h	evaluator non-sovereignty parameter; the rule $h < 1$ (never a helix variable)

Sources

Structural comparison table and A/B/Z assignment: en.wikipedia.org/wiki/A-DNA (after Dickerson 1992; Franklin 1953). Program context: peaice.org/eev3, peaice.org/ddatl, KakeyaLogic repository github.com/Manny536/kakeyalogic (PR #3), and lovelabslca.com. Primary structural literature as listed in Appendix A.

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